

Towards Smarter Clinical Predictions: Foundation Models for Integrating Multi-Source Domain Knowledge in EHRs

Jialun Wu, Kai He, *Member, IEEE*, Zeyu Gao, Xuequn Shang, Mengling Feng, *Senior Member, IEEE*

Abstract—Medication recommendation (MR) is a crucial clinical prediction task that supports physicians in prescribing effectively for patients with complex conditions. Effective MR requires integrating diverse medical knowledge and intricate relationships within clinical data. Traditional methods, however, largely focus on modeling relationships among medical entities and learning patient representations, often constrained by their reliance on single-domain knowledge. These constraints limit their ability to address the complexities of real-world scenarios. Recent advancements in foundation models present a promising solution by unifying multi-source data and uncovering rich semantic and structural insights. Yet, their potential in MR remains underexplored. To overcome these challenges, we introduce **MEDICS**, a foundation-model-driven framework that integrates multi-source domain knowledge and unifies semantic and structural information for enhanced clinical predictions. **MEDICS** consists of three synergistic modules: *KnowRetr* for multi-source medical knowledge retrieval, *MedChat* for LLM-based semantic reasoning, and *HyperMed* for high-order structural modeling using sub-hypergraphs. **MEDICS** employs a systematic process to retrieve, organize, and summarize domain knowledge for each medical code, utilizing this knowledge through two specialized modules. The *MedChat* module leverages large language models (LLMs) to redefine MR as a conversational question-answering task, enabling advanced semantic analysis. Simultaneously, the *HyperMed* module refines a hypergraph-based model by incorporating textual and structural embeddings to model high-order interactions among medical codes. Additionally, our collaborative training strategy further synergistically harnesses information from both modules. Extensive experiments on real-world EHR datasets (MIMIC-III and MIMIC-IV) demonstrate that **MEDICS** significantly outperforms state-of-the-art methods, setting a new benchmark for clinical prediction tasks.

Index Terms—Electronic health records, Data mining,

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Jialun Wu and Xuequn Shang are with the School of Computer Science, Northwestern Polytechnical University, 1 Dongxiang Rd, 710129, Xi'an, China. (E-mail: jialunwu.shang@nwpu.edu.cn).

Kai He and Mengling Feng are with the School of Saw Swee Hock School of Public Health, National University of Singapore. (E-mail: kaihe, ephfm@nus.edu.sg).

Zeyu Gao is with the CRUK Cambridge Centre, the Department of Oncology, University of Cambridge, United Kingdom. (E-mail: zg323@cam.ac.uk).

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I. INTRODUCTION

Electronic Health Records (EHRs) are critical to modern healthcare, serving as comprehensive repositories of patient information, including diagnoses, procedures, and medication histories [12], [13], [25], [38]. These records provide a rich foundation for developing advanced predictive models that enhance clinical decision-making tasks, such as medication recommendation (MR), which aids physicians in managing complex diseases [3], [4]. Recent advancements in foundation models, particularly in vision-language and language-only modalities, offer unprecedented opportunities to generalize across diverse healthcare data sources, making them well-suited for addressing the inherent complexity of EHR data [10], [44].

Building on the strengths of foundation models, recent studies have begun to dynamically utilize EHR data to improve prediction accuracy [11], [21]. For instance, medication prescriptions derived from patient visit data typically rely on instance-based [45] or longitudinal-based methods [34]. While instance-based approaches focus on isolated patient visits, longitudinal methods incorporate historical patient data, capturing long-term patterns that are essential for disease management. Moreover, studies integrating external biomedical knowledge into predictive tasks have shown promise in enhancing model performance [42], [43]. Foundation models, with their ability to unify and encode multi-source knowledge, provide a scalable and systematic approach to advancing these efforts, offering richer semantic understanding and a broader contextual framework for clinical predictions.

Despite these advancements, two critical challenges persist: (1) **Limited Multi-Source Knowledge Integration**. Current approaches often rely on single-source domain knowledge, such as hierarchical relationships of ICD diagnosis codes [26], atom-atom graph and molecular structures of medication codes [42], [43]. This narrow focus underutilizes the full semantic richness of medical concepts and overlooks the synergistic potential of integrating multi-source knowledge, a key factor for comprehensive patient modeling and accurate predictions. (2) **Neglect of Implicit Correlations**. Current MR systems often focus solely on encoding explicit EHR data, concentrating on pairwise relationships between medical codes and overlooking

the complex, implicit correlations among patient visits and between historical and current data [5]. Recognizing these correlations is vital for an in-depth modeling of EHR data, allowing for a more holistic approach to data encoding.

To address these challenges, we propose MEDICS, a multi-modal framework designed to advance clinical predictions. Leveraging the adaptability and generalization capabilities of foundational models, MEDICS systematically integrates multi-source domain knowledge and unifies semantic and structural learning. This integration enables the framework to effectively process diverse data modalities while emulating the clinical decision-making process. By synthesizing patients' historical records with data from current visits, MEDICS enhances prediction accuracy, providing a comprehensive and robust approach to complex healthcare scenarios. The framework achieves this through two core modules:

- 1) **Semantic learning module (MedChat):** This module utilizes large language models (LLMs), a class of foundation models, to transform MR tasks into conversational question-answering tasks. Tailored prompt templates refine structured EHR data into contextualized inputs for LLMs, leveraging their advanced semantic comprehension and multi-turn dialogue capabilities to enhance predictive accuracy.
- 2) **Structural learning module (HyperMed):** Built on sub-hypergraph neural networks, this module encodes domain knowledge into structural representations, capturing higher-order interactions between medical codes and patient visits. By integrating textual and structural data, HyperMed bridges the gap between semantic richness and structural complexity.

To unify these modules, we implement a collaborative training strategy that capitalizes on their synergistic potential, enhancing the framework's generalization ability and overall prediction accuracy. The contributions of our framework are summarized as follows:

- We propose MEDICS, a novel foundation-model-driven framework that integrates multi-source domain knowledge across both semantic and structural dimensions to address key limitations in EHR-based clinical predictions. Unlike prior models that rely on single-source or fixed-form knowledge representations, MEDICS systematically retrieves, organizes, and contextualizes knowledge from heterogeneous resources.
- We develop three complementary modules to support multi-level clinical modeling and emulate medical decision-making workflows: (1) KnowRetr integrates structured and unstructured knowledge sources to generate semantically enriched representations of medical codes, addressing limitations of prior single-source MR systems; (2) MedChat reformulates medication recommendation as a multi-turn question-answering task, leveraging LLMs for semantic reasoning beyond traditional encoder-based or graph-based models; (3) HyperMed utilizes hypergraph-based learning to capture high-order relationships and integrate semantic information into structural modeling, improving the cohesion of medical

codes and textual data. These modules are jointly optimized via a mutual-information-driven training objective, promoting semantic-structural alignment and enhancing robustness. Together, they simulate clinical reasoning and enable flexible integration into foundation-model-based medical applications.

- Comprehensive experiments on two EHR datasets demonstrate that MEDICS consistently outperforms existing predictive models, showcasing its flexibility as a plug-in architecture for foundational models and its potential to advance real-world healthcare applications.

II. RELATED WORK

A. Clinical Prediction Modeling

In recent years, clinical prediction systems have attracted substantial interest as they offer promising solutions for improving the accuracy and efficiency of medical predictions [27], [28]. These algorithms are generally divided into instance-based and longitudinal-based methods, based on their use of historical data. (1) *Instance-based methods*: These methods provide MR based solely on data from the current visit, such as diagnoses and procedures. For example, LEAP [45] identifies relationships between multiple diagnoses, treating the MR task as a series of decisions based on the current visit's data. However, these methods often fail to account for the fact that identical diagnoses can stem from different underlying causes, leading to diverse treatment needs. This limitation is primarily due to the absence of patient historical health records in the analysis. (2) *The longitudinal-based methods* have been developed to overcome this limitation. RETAIN [5] pioneered time-series predictions in healthcare. 4SDrug [36] utilizes symptoms as input and recommends medications by calculating the similarity between the patient symptoms and medication sets. G-Bert [33] incorporates pre-training techniques to enhance recommendations. Subsequent models like GAMENet [34] and MK-GNN [7] have enhanced predictions by incorporating advanced encoding techniques and medication co-occurrence graphs from EHRs. DrugRec [35] employs causal models to reduce recommendation bias, while some other methods have improved model performance by integrating external knowledge. SafeDrug [42], CSEDrug [39] and MoleRec [43] improve medication representations by integrating detailed atom-level and molecular-level information, addressing drug-drug interaction issues [23], [24]. Furthermore, some studies incorporate prescription history, which plays a crucial role in MR. For instance, MICRON [41] and COGNet [40] copy historical prescriptions into the current recommendation set with a certain probability, with COGNet directly inputting records into a Transformer encoder for modeling. Despite significant advancements, most existing methods primarily rely on EHR or medication-related knowledge, neglecting domain knowledge related to clinical manifestations and the medical semantics contained in EHRs.

In light of the above limitations, we propose MEDICS, a unified and modular framework that integrates external medical knowledge, semantic reasoning, and structural modeling to advance clinical prediction tasks. Our approach introduces three

interdependent innovations: (1) the integration of multi-source domain knowledge to enrich medical code representations beyond predefined ontologies; (2) the incorporation of foundation language models to support contextualized, interpretable reasoning over patient data; and (3) the construction of high-order relational structures that capture temporal and semantic dependencies across visits. These components are designed to function in a cohesive manner, forming a clinically motivated architecture that supports both accuracy and generalizability. We describe the components of this framework in the next section.

B. Knowledge Sources Descriptions

DrugBank¹: DrugBank is a comprehensive database offering detailed information on drugs and their targets, including chemical structures, pharmacological profiles, and molecular pathways. It contains over 355,000 entries and serves as a key resource in pharmacology and clinical research by supporting tasks such as drug discovery and mechanism interpretation.

Wikipedia²: Wikipedia is a multilingual, collaborative encyclopedia maintained by the Wikimedia Foundation. For medical research, it provides a broad range of healthcare-related articles. Using lists of standardized medical codes (e.g., ICD, ATC), approximately 150,000 relevant entries can be extracted to support knowledge retrieval and background understanding in clinical domains.

International Classification of Diseases (ICD)³: Maintained by the World Health Organization, ICD is a globally adopted system that encodes diseases, symptoms, and external causes using standardized codes. It supports diagnostic consistency, health statistics, and epidemiological research across healthcare systems.

Anatomical Therapeutic Chemical (ATC)⁴: Also issued by the WHO, the ATC system classifies drugs based on the organ/system targeted and their therapeutic, pharmacological, and chemical characteristics. It follows a five-level hierarchy and is widely used for drug utilization analysis, inventory management, and cross-national pharmacoepidemiological studies.

III. METHOD

In this section, we first formulate the problem setup and outline the structure of our proposed framework, MEDICS.

A. Problem Setup

EHRs Data: The longitudinal EHRs data includes a cohort of patients, denoted by N , with each represented by a sequence of hospital visits $X_i = \{x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(T_i)}\}$. Each visit x_i includes a set of medical codes $C_i \subseteq C$. Specifically, the t -th visit of patient i is characterized by an array of diagnoses, procedures, and medications, denoted as $x_i^{(t)} = [d_i^{(t)}, p_i^{(t)}, m_i^{(t)}]$. Each medical code $c_i \in C_i$ signifies a specific clinical event. The important notations and its description utilized in this paper are listed in Table I.

TABLE I
THE UTILIZED NOTATIONS IN MEDICS.

Notation	Description
$x_i^{(1)}, \dots, x_i^{(T_i)}$	patient i 's health states X_i in each visit
$d_i^{(t)}, p_i^{(t)}, m_i^{(t)}$	patient i 's clinical visit record at t -th visit $v_i^{(t)}$
D, P, M	Diagnosis, Procedure, and Medication set
c_i	the medical code
$k_{i,1}, \dots, k_{i,m}$	the retrieved knowledge paragraph
s_i	the summarized knowledge for medical code c_i
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	the constructed patient hypergraph
e_j, v_i	the hyperedge, node embedding
y_1, y_2	the prediction result in MedChat and HyperMed
$\hat{y}$	the predict prescriptions (compare to δ)



Fig. 1. High-level overview of the MEDICS framework.

Problem Definition: Given the clinical record x_i with the associated medical codes C_i , the core objective is to develop a predictive function $f(\cdot)$ capable of accurately predicting a patient's clinical outcomes y_i at each visit t . The prediction result $\hat{y}^{(t)}$ should closely align with the actual clinical event $y^{(t)}$. By leveraging extensive longitudinal data, our framework aims at conducting a comprehensive analysis of each patient's historical data to enhance the accuracy of clinical outcome predictions.

B. Proposed Framework

Structured EHR data is typically organized chronologically by patient visits and stored in a tabular format using standardized medical coding. Most existing approaches generally represent each patient visit as a flat set of these medical codes, essentially treating them as context-free vectors. These methods often neglecting the rich semantic content or the intrinsic relationships among them. Moreover, with the increasing availability of multi-modal EHR data and external medical knowledge bases, there is an untapped opportunity to enrich code representations using multi-source semantics.

To address these limitations and simulate physician-like diagnostic reasoning, MEDICS introduces a modular architecture comprising three interrelated components (Figure 1):

- **KnowRetr** systematically retrieves, organizes, and summarizes relevant domain knowledge for each medical code, providing semantically enriched representations that serve as external context.
- **MedChat** formulates MR as a conversational multi-turn question-answering task and leverages a large language model to reason over patient history, thereby enhancing semantic analysis and interpretation.
- **HyperMed** constructs patient-specific sub-hypergraphs across visits and incorporates textual semantics into node embeddings, enabling high-order structural modeling of medical code interactions.

¹<https://go.drugbank.com/releases/latest>

²<https://www.wikipedia.org/>

³<https://www.who.int/standards/classifications/classification-of-diseases>

⁴<https://www.who.int/tools/atc-ddd-toolkit/atc-classification>

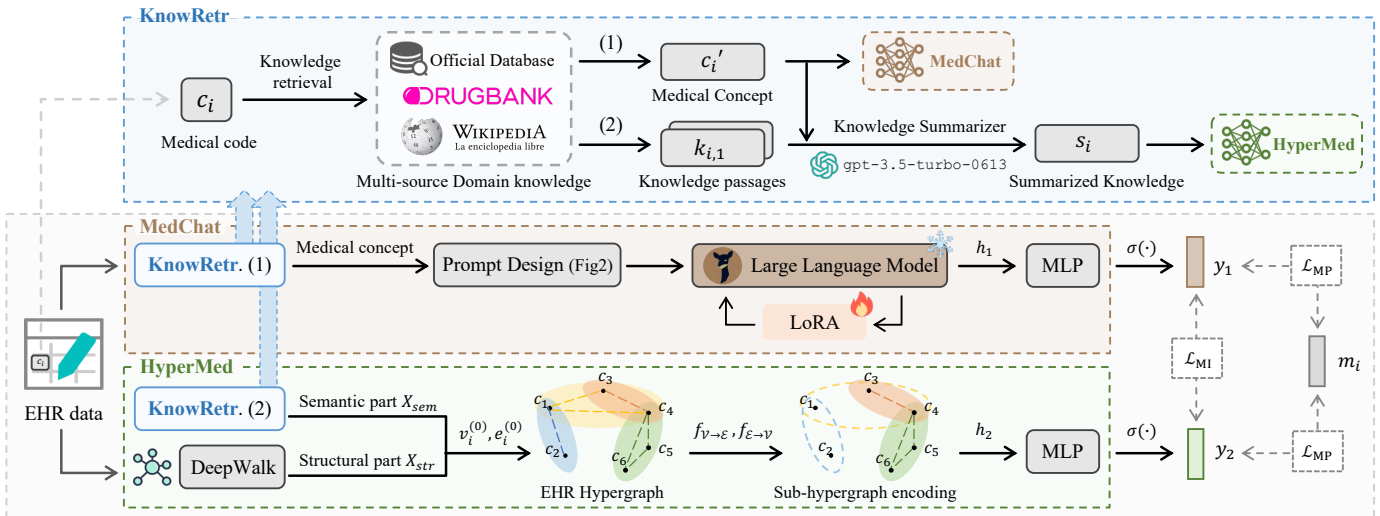


Fig. 2. An overview of MEDICS framework. The input of MEDICS are the patient tabular EHRs data and multi-source domain knowledge. The output of MEDICS is the predicted medication combination. MEDICS consists of three modules, (a) KnowRetr, which retrieves and summarizes domain knowledge for each medical code. (b) MedChat, which redefines MR into conversational QA and uses LLMs to encode semantic information. and (c) HyperMed, which enhances hypergraphs with fusion embedding to analyze complex correlations.

These modules are jointly trained under a mutual-information-based objective that encourages semantic-structural alignment and robust clinical prediction. This architecture enables flexible, interpretable, and clinically aligned decision-making, while supporting plug-in compatibility with other foundation-model-driven applications. Each module is elaborated below in turn.

1) KnowRetr: To enhance the utility of medical codes by enriching them with comprehensive semantic information, we designed a knowledge retrieval enhancement module. This module is tasked with sourcing and aligning the most relevant medical concepts and definitions for each medical code, comprising two main steps: Domain Knowledge Retrieval and Knowledge Summary Generation.

(1) Domain Knowledge Retrieval. Initially, our module accesses multiple authoritative online databases including the International Classification of Diseases (ICD), Anatomical Therapeutic Chemical Classification System (ATC), Wikipedia, and DrugBank [37]. These sources are chosen for their rich and accurate medical content, ensuring that the medical codes are supplemented with corresponding concepts and comprehensive multi-source medical descriptions. This retrieval process is foundational, providing the critical data required for the effective operation of the subsequent MedChat module.

(2) Knowledge Summary Generation. To effectively distill and condense the multi-source descriptive information, we harness the advanced text summarization capabilities of large language models (LLMs) such as gpt-3.5-turbo-0613. The process generates summarized knowledge s_i for each medical code c_i as:

$$s_i = \text{LLM}([\text{Prompt}, k_{i,1}, \dots, k_{i,m}]), \quad (1)$$

where $k_{i,m}$ represents the retrieved knowledge passages.

By embedding downstream task-related information into the designed prompts, we ensure that the generated summaries

are both concise and highly relevant to the specific medical context. We present the detailed design of the prompt template as follows:

LLM Summarization Task
Imagine you are a physician tasked with predicting health-related outcomes. You need specific information for a given <task>. Here is the provided information: - Type: <Medical Code Type> - Name: <Medical Code Name> - Retrieve Knowledge Passage #1: <Passage 1> - Retrieve Knowledge Passage #2: <Passage 2> ... Based on the information provided, can you generate a concise summary (10-20 words) that encapsulates the essential knowledge of the <Medical Code Name> for the specified <task>?

This tailored summary s_i is then integrated into the HyperMed module for further analysis. To construct the prompt, we adopted a manual and iterative design strategy. Specifically, the prompt was refined through a series of internal trials across different medical codes, with the goal of producing summaries that are concise, clinically relevant, and informative for downstream tasks. Domain researchers qualitatively assessed the outputs to ensure alignment with clinical semantics and usability. This empirical, task-oriented approach allowed us to achieve consistent and relevant summarization performance within the medication recommendation setting.

To boost efficiency, we index the entire corpus prior to its application in prediction tasks and cache the summarized knowledge for each medical code, thereby minimizing redundant operations. This approach minimizes redundant operations and enhances the speed of information retrieval. Through these steps, KnowRetr systematically augments the medical codes with enriched semantic information.

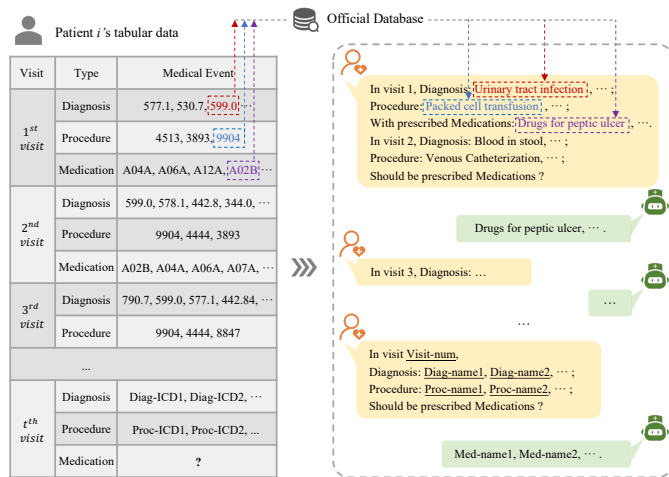


Fig. 3. Example of temporal conversations construction. Underlined sections are filled with EHR data. Visit-num indicates hospital visits. Standard medical concepts from KnowRetr are used for diagnoses, procedures, and medications (Diag-name, Proc-name, and Med-name), allowing LLMs to understand MR from the input.

2) MedChat: In real clinical settings, physicians assess a patient's current diagnosis and examination results with historical clinical records to make accurate clinical decisions. To simulate this process, we redefine the MR task as a conversational question-answering format and developed MedChat, a chatbot capable of emulating the diagnostic capabilities of clinical physicians. As illustrated in Figure 3, we designed prompt templates for patient visit records. These templates transform structured medical records—enriched with domain knowledge—into a series of questions, with prescriptions serving as answers. This setup fosters temporal conversations that encapsulate the clinical context, which are then input into a LLM for fine-tuning. This approach allows the LLM to synthesize current diagnostic results with relevant historical data, thereby facilitating the generation of accurate medication prescriptions.

Traditional LLM-based recommendation systems output recommendations in natural language form but struggle with the out-of-corpus challenge. To overcome this, we replaced the pre-trained word generation layer with a linear layer equipped with a sigmoid function, allowing the model to output a probability distribution for each medication:

$$y_1 = \sigma(W_1 \cdot h_1), \quad (2)$$

where y_1 is the predicted probability of medication, h_1 denotes the hidden states from the last transformer layer of the LLM, W_1 is a learnable weight matrix, and $\sigma(\cdot)$ is the sigmoid function.

Given the extensive number of parameters in LLMs, full-parameter fine-tuning can be computationally expensive. To address this, we employed LoRA (Low-Rank Adaptation) [14], a lightweight fine-tuning technique that inserts trainable low-rank matrices into each layer of the model, allowing efficient training with fewer updated parameters. This adaptation enables the modified LLM to output probabilities over a set of

medications rather than predicting exact drug names directly.

MedChat enhances the understanding and management of complex clinical information across three key dimensions: • *Disease Progression*: MedChat's time-series analysis capabilities allow it to analyze correlations between current and historical health states, track disease progression, and identify key changes in indicators. • *Diagnostic Process*: Leveraging its generalization of medical knowledge, MedChat simulates a physician's diagnostic logic to form diagnostic insights. • *Treatment Recommendations*: MedChat assesses treatment effectiveness based on the evolution of health status (medical and prescription history) and recommends the most appropriate medication accordingly.

3) HyperMed: In the realm of modeling structured EHR data, hypergraphs offer a sophisticated method to capture complex relationships between visits and medical codes, proving superior to conventional graph structures [1], [2]. Hypergraphs convert tabular EHR data by treating each medical code as a node and each visit as a hyperedge. Each hyperedge includes all nodes relevant to a specific visit. The hypergraph is denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_i\}_{i=1}^{|C|}$ and $\mathcal{E} = \{e_i^{(t)}\}$ denote the nodes and hyperedges, with all nodes being elements of the set of medical codes C .

As shown in the bottom part of Figure 2, initializing node embeddings for medical codes in a hypergraph involves both structural and semantic aspects. *Structurally*, we utilize the DeepWalk algorithm [31], which performs random walks on hypergraph $\mathcal{G}$ to train a Skip-gram model. This method effectively captures the each node embedding by predicting contextual neighbors, thereby generating the structural embeddings X_{str} . *Semantically*, we apply PubmedBERT [9], a Transformer-based model pretrained on extensive biomedical texts literature, to process summarized medical code descriptions e_i from the KnowRetr module, resulting in semantic embeddings X_{sem} . Although our task is clinical prediction, the semantic content encoded here is not raw EHR text, but rather structured descriptions drawn from ICD, ATC, DrugBank, and Wikipedia. These sources are formal and biomedical in nature, and thus more closely aligned with PubMedBERT's training corpus than with the clinical notes used to train models such as ClinicalBERT. We found that concatenating these embeddings initially offers optimal performance:

$$v_i^{(0)} = [X_{\text{str}}; X_{\text{sem}}]. \quad (3)$$

The critical step in linking node embeddings with MR involves optimizing the use of visit-level information and enhancing the representation of hyperedges. Existing methods, such as the HGNN [8] and Hypergraph Transformer [22], inadequately capture the breadth of a patient's health status, focusing narrowly on single visits.

To address this, the HyperMed module improves patient health depiction via a two-step approach. It first constructs a *sub-hypergraph* within $\mathcal{G}$ by amalgamating hyperedges from both current and previous visits, thus capturing a comprehensive patient history. It then encodes each sub-hypergraph to more accurately reflect the patient's health state. The initial embedding for each hyperedge, $e_j^{(0)}$, is derived by averaging

the embeddings $v_i^{(0)}$ associated with that hyperedge. In layers $\ell \in (1, \ell_s)$, node and hyperedge embeddings are iteratively updated through a two-step message-passing mechanism:

$$e_j^{(\ell)} = f_{\mathcal{V} \rightarrow \mathcal{E}}(e_j^{(\ell-1)}, \mathcal{V}^{(\ell)}), \mathcal{V}^{(\ell)} = \{v_i^{(\ell)} \in e_j\}, \quad (4)$$

$$v_i^{(\ell+1)} = f_{\mathcal{E} \rightarrow \mathcal{V}}(v_i^{(\ell)}, \mathcal{E}^{(\ell)}), \mathcal{E}^{(\ell)} = \{e_j^{(\ell)} \ni v_i\}. \quad (5)$$

Initially, the hyperedge embeddings transition from $e_j^{(\ell-1)}$ to $e_j^{(\ell)}$, incorporating information from all node $\mathcal{V}^{(\ell)}$ within the hyperedge e_j . Subsequently, node embeddings update from $v_i^{(\ell)}$ to $v_i^{(\ell+1)}$, integrating data from all hyperedges $\mathcal{E}^{(\ell)}$ that contain node v_i . To implement the update functions $f(\cdot)$, we utilize a cross-attention mechanism, which computes how strongly each node and hyperedge pair influences one another. Specifically, for two different types of entities (e.g., nodes and hyperedges), the mechanism assigns attention weights based on their interaction scores, enabling selective information exchange:

$$\mathcal{Z}^{(\ell+1)} = \sum_{\mathcal{E}, \mathcal{V}} a(\mathcal{Z}^{(\ell)}, \mathcal{Y}^{(\ell)}) \mathcal{Y}^{(\ell)}, \quad (6)$$

$$a(\mathcal{Z}^{(\ell)}, \mathcal{Y}^{(\ell)}) = \frac{\exp(s(\mathcal{Z}^{(\ell)}, \mathcal{Y}^{(\ell)}))}{\sum_{\mathcal{E}, \mathcal{V}} \exp(s(\mathcal{Z}^{(\ell)}, \mathcal{Y}^{(\ell)}))}, \quad (7)$$

where $s(\mathcal{Z}, \mathcal{Y})$ is a scoring function that evaluates the similarity or association between two entities:

$$s(\mathcal{Z}, \mathcal{Y}) = (W_v \mathcal{Z} + b_v) \odot (W_e \mathcal{Y} + b_e), \quad (8)$$

with W_v , W_e , b_v , and b_e are trainable parameters. HyperMed effectively leverages the updated hyperedge to enhance node embeddings, capitalizing on the unique advantages of hypergraphs to capture complex relationships.

After progressing through ℓ_s layers, HyperMed integrates the hyperedges of the sub-hypergraph in the final layer using a weighted attention mechanism:

$$h_2 = \sum_{e_i \in \mathcal{G}_j} \frac{\exp(w_e^T e_i^{(\ell_s)}) e_i^{(\ell_s)}}{\sum_{e_{i'} \in \mathcal{G}_j} \exp(w_e^T e_{i'}^{(\ell_s)})}, \quad (9)$$

where w_e is a learnable parameter. This approach ensures a comprehensive representation that encapsulates the intricate relationships and interactions within the sub-hypergraph, reflecting the patient's current health state accurately. To further parameterize h_2 , we use an MLP layer to generate the final medication combinations:

$$y_2 = \sigma(W_2 \cdot h_2), \quad (10)$$

where W_2 denotes the learnable parameters, y_2 denotes the final matching scores for a patient.

4) Co-training: Using the MedChat and HyperMed modules, we obtain two distinct predictive values, y_1 and y_2 , respectively. With the collaborative training strategy, we define the learning objectives as follows:

$$\mathcal{L}_{mc} = \mathcal{L}_{mp}(y_1, m_i) + \lambda \mathcal{L}_{MI}(y_1, y_2), \quad (11)$$

$$\mathcal{L}_{hm} = \mathcal{L}_{mp}(y_2, m_i) + \lambda \mathcal{L}_{MI}(y_2, y_1), \quad (12)$$

where m_i denotes target prescriptions, $\mathcal{L}_{mc}$ and $\mathcal{L}_{hm}$ are reciprocal losses tailored to regulate and stabilize the training of both modules. The $\mathcal{L}_{mp}(\cdot)$ represents the multi-label prediction

loss, formulated by integrating the Multi-Label Hinge Loss ($\mathcal{L}_{mlh}$) [17] and the Binary Cross-Entropy Loss ($\mathcal{L}_{bce}$) through a balance parameter μ :

$$\mathcal{L}_{mp} = \mu \mathcal{L}_{mlh} + (1 - \mu) \mathcal{L}_{bce}. \quad (13)$$

The $\mathcal{L}_{MI}(\cdot)$ quantifies the mutual information between the predictions y_1 and y_2 , optimized using the InfoNCE loss function. This optimization aims to maximize mutual information, thus exploiting data from other modules to bridge any information gaps, ensuring consistent representations across modules, and enhancing framework robustness.

The combined prediction y integrates the outputs from both predictors using a parameter β :

$$y = \beta y_1 + (1 - \beta) y_2. \quad (14)$$

During the inference stage, y is directly used as the final prediction. By comparing y against a predefined threshold parameter δ , the final medication combinations $\hat{y}$ are determined by our framework. The overall algorithm of the proposed framework is detailed in the supplementary material.

IV. EVALUATION

In this section, we detail the extensive experiments conducted to address the following pivotal research questions (RQs):

- **RQ1:** Does MEDICS outperform state-of-the-art methods in providing accurate medication recommendations?
- **RQ2:** Are the individual modules within MEDICS effective for medication recommendation tasks?
- **RQ3:** Does the integration of MEDICS modules enhance the accuracy of existing methods consistently?
- **RQ4:** How does the amount of previous patient visits utilized affect the performance of MEDICS?

A. Experiment Setups

1) Datasets: Our experiments employed two widely available real-world EHR datasets: MIMIC-III and MIMIC-IV [18], [19]. MIMIC-III encompasses data collected between 2001 and 2012, whereas MIMIC-IV includes records spanning from 2008 to 2019. These datasets play a critical role in advancing medical research by providing real-world clinical data for diverse analyses, modeling efforts, and studies focused on improving patient care and health outcomes. For data processing, diagnosis, procedure, and medication records were extracted from each patient visit, utilizing various clinical tables in alignment with prior research [35]. To ensure the inclusion of comprehensive historical health information, we restricted the analysis to patients with multiple visits, focusing on medication records from the first 24 hours of each visit. Table II outlines detailed statistics for the MIMIC-III and MIMIC-IV datasets utilized in this study [18], [19].

2) Baselines: We assess the performance of MEDICS by comparing it with the following baseline methods, including (1) Instance-based methods: Logistic Regression (LR), Ensembled Classifier Chain (ECC) [32], LEAP [45], and (2) Longitudinal-based methods: RETAIN [5], G-BERT [33],

TABLE II
THE STATISTICS FOR MIMIC-III AND MIMIC-IV.

Statistics	MIMIC-III	MIMIC-IV
# of patients	5,208	6,136
# of clinical events	12,490	17,813
# of diagnoses	1,895	1,851
# of procedures	1,378	4,001
# of medications	112	121
# of visits per patient (avg.)	2.59	2.90
# of diagnoses per visit (avg.)	10.24	11.78
# of procedures per visit (avg.)	3.85	2.18
# of medications per visit (avg.)	11.30	6.68

GAMENet [34], SafeDrug [42], MICRON [41], COGNet [40], and DrugRec [35].

The following provides a brief description of these baseline methods:

- **Logistic Regression (LR)** [16]: A traditional instance-based linear classifier with $L2$ regularization for prediction.
- **Ensemble Classifier Chain (ECC)** [32]: A multi-label model that sequentially trains classifiers, with each classifier leveraging previous predictions as input features.
- **LEAP** [45]: An instance-based sequential decision framework using a recurrent decoder to model label dependencies and content-based attention to associate labels with instances.
- **RETAIN** [5]: A sequence-based method employing a two-level RNN with attention to identify critical past visits and relevant clinical variables.
- **G-BERT** [35]: A pre-trained model on diagnosis and medication data, requiring historical medication records during fine-tuning.
- **GAMENet** [34]: Combines memory-augmented neural networks and Drug-Drug Interaction graphs to capture medication history and interactions.
- **SafeDrug** [42]: Integrates global and local encoders to incorporate molecular structure data for medication recommendation.
- **MICRON** [41]: A recurrent residual learning approach focusing on incremental changes in medications and diseases.
- **COGNet** [40]: Uses a Transformer-based copy-or-predict mechanism for generating medication combinations based on past prescriptions.
- **DrugRec** [35]: Addresses recommendation bias using causal graph modeling to identify confounding factors.

3) **Evaluation Metrics**: Building on previous research, the performance of MEDICS in medication recommendation is evaluated through a combination of accuracy, effectiveness, and safety metrics. These metrics include:

- **Jaccard Similarity Score (Jaccard)**: This score quantifies the similarity and diversity between predicted and actual sets of medications [30]. It is calculated as:

$$\text{Jaccard}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} \frac{|\{y_{i,j}^{(t)} = 1\} \cap \{\hat{y}_{i,j}^{(t)} = 1\}|}{|\{y_{i,j}^{(t)} = 1\} \cup \{\hat{y}_{i,j}^{(t)} = 1\}|}. \quad (15)$$

- **Precision-Recall Area Under the Curve (PRAUC)**: PRAUC measures the balance between precision and recall over varying thresholds [6], defined as:

$$\text{PRAUC}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} \sum_{k=1}^{|M|} P(k)_i^{(t)} (R(k)_i^{(t)} - R(k-1)_i^{(t)}), \quad (16)$$

where k represents the rank within the sequence of retrieved medications.

- **Average F1 Score (F1)**: F1 combines precision and recall into a single metric, capturing their harmonic mean:

$$\text{F1}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} \frac{2P_i^{(t)}R_i^{(t)}}{P_i^{(t)} + R_i^{(t)}}, \quad (17)$$

where Precision $P_i^{(t)}$ and Recall $R_i^{(t)}$ are defined as:

$$P_i^{(t)} = \frac{|\{m_{i,j}^{(t)} = 1\} \cap \{\hat{y}_{i,j}^{(t)} = 1\}|}{|\{\hat{y}_{i,j}^{(t)} = 1\}|}, \quad (18)$$

$$R_i^{(t)} = \frac{|\{m_{i,j}^{(t)} = 1\} \cap \{\hat{y}_{i,j}^{(t)} = 1\}|}{|\{m_{i,j}^{(t)} = 1\}|}. \quad (19)$$

- **Drug-Drug Interaction (DDI) Rate**: The DDI rate, reflecting the safety of medication combinations, is calculated for patient i as:

$$\text{DDI}_i = \frac{\sum_{t=1}^{T_i} \sum_{k,l \in \{\hat{y}_{i,j}^{(t)} = 1\}} \mathbf{1}\{\mathcal{M}_{kl} = 1\}}{\sum_{t=1}^{T_i} \sum_{k,l \in \{\hat{y}_{i,j}^{(t)} = 1\}} \mathbf{1}}. \quad (20)$$

All metrics are macro-averaged across patients to provide a comprehensive evaluation across dimensions, ensuring a holistic understanding of the framework's performance.

To ensure the robustness of our experimental results, we employed bootstrapping sampling during the testing process. Specifically, we randomly sampled 80% of the data in each round, and the metrics reported are the averages from ten rounds of testing. As presented in Tables III and IV, and Figure 4, all improvements are statistically significant with a p -value of less than 0.05.

4) **Implementation Details**: To ensure a fair comparison, we conduct most of our experiments using PyTorch and set the representation dimensionality to 64 for all methods. For the baselines, we adhere to the original settings recommended by the authors. In the medication recommendation task, we split the dataset into training, validation, and testing sets in a 4:1:1 ratio, consistent with previous work [35]. All models in our experiments are optimized using the Adam optimizer [20]. We perform fine-tuning on the Vicuna-13b model using the FastChat platform, which facilitates the training, deployment, and evaluation of LLM chatbots. Vicuna's advanced multi-turn dialogue capabilities and robust performance in various NLP tasks, enhanced through LoRA fine-tuning, make it a powerful tool for interactive dialogue systems. The fine-tuning configurations include a LoRA rank of 16, a batch size of 32, a learning rate of 2×10^{-5} , and a maximum input length of 2,048 tokens. All experiments are conducted on a Linux server equipped with Tesla V100 32GB GPUs.

TABLE III
THE RESULTS OF COMPARATIVE EXPERIMENTS ON THE MIMIC-III AND MIMIC-IV DATASETS.

Model	MIMIC-III				MIMIC-IV			
	Jaccard	F1	PRAUC	DDI	Jaccard	F1	PRAUC	DDI
LR	48.96 ± 0.25	64.91 ± 0.24	75.68 ± 0.25	7.74 ± 0.12	38.44 ± 0.28	53.79 ± 0.31	<u>65.68</u> ± 0.36	6.45 ± 0.12
ECC	47.99 ± 0.22	63.90 ± 0.22	75.72 ± 0.26	7.60 ± 0.10	36.90 ± 0.41	51.73 ± 0.47	65.41 ± 0.30	6.48 ± 0.18
LEAP	44.65 ± 0.37	60.97 ± 0.36	64.90 ± 0.33	6.57 ± 0.10	36.53 ± 0.28	52.01 ± 0.33	53.14 ± 0.38	5.70 ± 0.11
RETAIN	47.80 ± 0.36	63.97 ± 0.36	76.01 ± 0.35	8.14 ± 0.18	39.03 ± 0.38	54.71 ± 0.40	65.63 ± 0.55	6.18 ± 0.25
G-BERT	46.51 ± 0.16	62.67 ± 0.15	75.10 ± 0.17	8.11 ± 0.11	36.93 ± 0.16	52.36 ± 0.19	62.66 ± 0.21	8.11 ± 0.11
GAMENet	50.39 ± 0.21	66.09 ± 0.20	76.32 ± 0.27	8.32 ± 0.05	39.57 ± 0.35	55.25 ± 0.41	64.79 ± 0.55	7.57 ± 0.14
SafeDrug	50.90 ± 0.38	66.64 ± 0.33	76.47 ± 0.20	6.58 ± 0.03	40.82 ± 0.26	56.51 ± 0.28	64.95 ± 0.36	<u>5.53</u> ± 0.10
MICRON	50.76 ± 0.37	66.34 ± 0.35	76.85 ± 0.38	<u>6.12</u> ± 0.08	40.09 ± 0.44	55.45 ± 0.48	65.84 ± 0.43	6.05 ± 0.17
COGNet	51.34 ± 0.27	67.06 ± 0.43	76.77 ± 0.13	7.84 ± 0.05	41.31 ± 0.20	56.60 ± 0.19	64.60 ± 0.17	5.96 ± 0.05
DrugRec	<u>52.20</u> ± 0.34	<u>67.71</u> ± 0.31	<u>77.20</u> ± 0.36	5.97 ± 0.06	<u>41.94</u> ± 0.20	<u>57.13</u> ± 0.22	65.58 ± 0.26	3.96 ± 0.07
MEDICS	53.68 ± 0.19	68.94 ± 0.14	79.03 ± 0.17	7.13 ± 0.12	43.48 ± 0.17	59.54 ± 0.21	67.35 ± 0.19	6.47 ± 0.08

Bold numbers indicate the best results, and underlined numbers indicate the second-best results in each category.

TABLE IV
ABLATION STUDIES ON THE MIMIC-III AND MIMIC-IV DATASETS.

Model					MIMIC-III				MIMIC-IV			
A	B	C	D	E	Jaccard	F1	PRAUC	DDI	Jaccard	F1	PRAUC	DDI
✓					52.87 ± 0.16	67.86 ± 0.17	78.53 ± 0.15	8.13 ± 0.11	41.15 ± 0.16	57.13 ± 0.18	66.85 ± 0.21	7.85 ± 0.07
	✓				53.16 ± 0.18	68.08 ± 0.21	78.85 ± 0.18	7.98 ± 0.09	41.45 ± 0.20	57.44 ± 0.19	66.67 ± 0.18	7.46 ± 0.10
		✓			53.02 ± 0.17	67.95 ± 0.16	78.91 ± 0.20	7.68 ± 0.08	41.32 ± 0.15	57.32 ± 0.23	66.49 ± 0.20	7.27 ± 0.09
			✓		51.47 ± 0.21	66.58 ± 0.23	78.27 ± 0.19	7.57 ± 0.10	40.84 ± 0.18	56.79 ± 0.22	65.83 ± 0.17	6.92 ± 0.14
				✓	53.52 ± 0.20	68.36 ± 0.21	78.75 ± 0.25	7.45 ± 0.07	41.83 ± 0.23	57.88 ± 0.18	66.54 ± 0.19	7.21 ± 0.11
✓	✓	✓	✓	✓	53.18 ± 0.19	68.94 ± 0.14	79.03 ± 0.17	7.13 ± 0.12	43.48 ± 0.17	59.54 ± 0.21	67.35 ± 0.19	6.47 ± 0.08

Component A refers to MedChat, B means HyperMed, and C represents Structure embedding, D represents Semantic embedding, E represents Co-training.

B. Results and Analysis

1) *RQ1: Overall Performance*: To demonstrate the efficacy of MEDICS, we evaluated its performance against existing baseline models, with results summarized in Table III. Across all assessed metrics, MEDICS consistently outperformed other models on two distinct datasets. For a more intuitive comparison, Figure 4 visually represents these metrics for each dataset in two subfigures. In these radar charts, each point signifies a specific performance metric, with points closer to the perimeter indicating better performance. Initially, MEDICS surpassed traditional models that do not utilize external knowledge (e.g., LR, ECC, LEAP, and RETAIN). This enhancement highlights the advantages of integrating domain knowledge into predictive models. Furthermore, MEDICS also outperformed advanced models that do incorporate external knowledge (e.g., GAMENet, SafeDrug, MICRON), due to its training approach that synthesizes multiple sources of external knowledge along with structural and semantic insights. Moreover, our analysis suggests that conventional methods using recurrent neural networks and their variants often produce suboptimal outcomes in medical records analysis, likely due to the inclusion of irrelevant data. In contrast, hypergraph modeling, which adeptly represents complex interrelationships within medical events and integrates summarized knowledge, has proven more effective for prediction tasks.

The comparative figure reveal that MEDICS typically excels most models in reducing the DDI rate across datasets, although it may not perform as well as some models that explicitly focus on minimizing DDIs through their loss functions. It's essential to recognize the trade-off between achieving accurate medication predictions and reducing DDI rates, crucial for

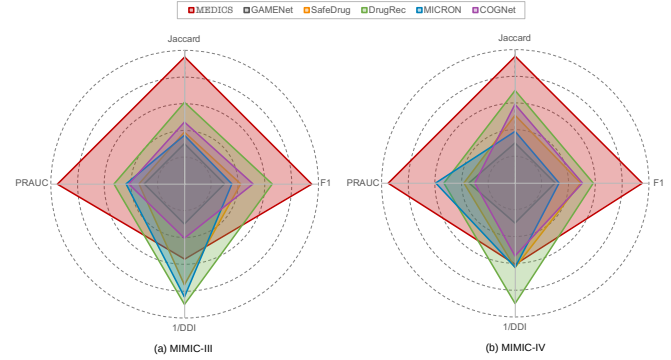


Fig. 4. Comparison with recent works across all metrics.

minimizing adverse drug reactions. For example, the average DDI rates in the two datasets are 0.0754 and 0.0713, respectively. Particularly in the MIMIC dataset, which targets ICU patients with complex medical conditions, DDIs are sometimes unavoidable to ensure effective treatment. Consequently, the closer the recommendations align with optimal treatment protocols, the closer the DDI rate may mirror the actual conditions of patient care. Despite these challenges, MEDICS effectively recommends the most appropriate medications while ensuring safety, demonstrating the capability to optimize treatment outcomes while adhering to crucial safety standards.

2) *RQ2: Ablation Study*: To assess the contribution of each component within MEDICS, we conducted experiments on MEDICS and its variants, as detailed in Table IV. The results demonstrate that the removal of any component significantly degrades performance, confirming the essential nature of our design. Using only the MedChat or the HyperMed module

TABLE V
CASE STUDY OF MEDICATION RECOMMENDATIONS FOR PATIENT WITH MULTIPLE VISITS.

Visit	Method	Recommended Medication Combination	Unseen & Missed Medication
1 st Visit	Ground-truth	N02B, A06A, B05C, C07A, C03C, B01A, N05C, C01A, C03D, M04A	
	GAMENet	N02B, A06A, B05C, C07A, C03C, B01A, N05C, C03D	+ 20 unseen + 2 missed
	SafeDrug	N02B, A06A, B05C, C07A, C03C, B01A, N05C, C03D	+ 11 unseen + 2 missed
	COGNet	A06A, B05C, C07A, C03C, B01A, N05C, C01A, C03D	+ 17 unseen + 2 missed
	MEDICS	N02B, A06A, B05C, C07A, C03C, B01A, C03D, M04A	+ 11 unseen + 2 missed
2 nd Visit	Ground-truth	N02B, A01A, A02B, A06A, B05C, A12C, C01C, A07A, C07A, C03C, A12B, B01A, N05C, J01D, R06A, N03A, N05A, A04A, R03A, N05B, R05C, D06A, R01A, C03D, M04A	
	GAMENet	N02B, A01A, A02B, A06A, B05C, A12C, C01C, A07A, C07A, C03C, A12B, B01A, N05C, J01D, N05A, A04A, R03A, N05B, R01A, C03D	+ 10 unseen + 5 missed
	SafeDrug	N02B, A01A, A02B, A06A, B05C, A12C, C01C, A07A, C07A, C03C, A12B, B01A, N05C, J01D, N05A, A04A, R03A, N05B, C03D	+ 8 unseen + 6 missed
	COGNet	N02B, A01A, A02B, A06A, B05C, A12C, C01C, A07A, C07A, C03C, A12B, B01A, N05C, J01D, N05A, A04A, R03A, R01A, C03D, M04A	+ 8 unseen + 5 missed
	MEDICS	N02B, A01A, A02B, A06A, B05C, A12C, C01C, A07A, C07A, C03C, A12B, B01A, N05C, J01D, N05A, A04A, R03A, R05C, R01A, C03D, M04A	+ 7 unseen + 4 missed

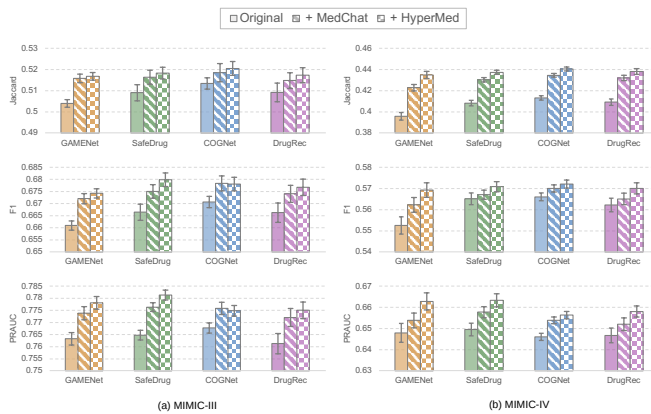


Fig. 5. Performance of three longitudinal-based methods and their variants equipped with two modules on two datasets.

yielded suboptimal results. Specifically, removing the HyperMed module disrupts the ability to capture complex correlations among medical codes, and removing the MedChat module results in the loss of crucial semantic information. Further ablation studies focused on the processing of semantic and structural data within the HyperMed module. These studies showed that excluding semantic information markedly reduces performance across all four metrics, highlighting the critical role of clinical concepts and summarized knowledge in providing structured modeling data. Additionally, eliminating the co-training strategy, which integrates the representations from the two modules for final predictions, also results in poorer performance. This finding emphasizes the importance of co-training in enhancing model robustness, leveraging comprehensive information, and improving the stability of the model.

3) RQ3: Compatibility of Core Modules: We propose that the two primary modules of MEDICS (*i.e.*, MedChat and HyperMed) can significantly enhance the accuracy of various MR models when applied orthogonally. To substantiate this claim, we evaluated the performance enhancements of three longitudinally-based MR methods (*i.e.*, GAMENet, SafeDrug, and COGNet) alongside their variants that incorporate either

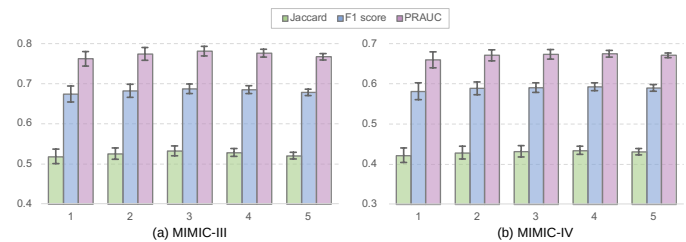


Fig. 6. The effect of the number of utilized historical visits.

the MedChat or HyperMed module. The findings, presented in Figure 5, indicate that the adapted models modified with our components consistently outperformed their standard configurations across the Jaccard, F1, and PRAUC metrics. The selected models for evaluation were chosen based on their diverse architectural foundations: GAMENet and SafeDrug are built on RNNs, whereas COGNet employs a Transformer architecture. This adaptability confirms the broad applicability and efficacy of our modules, underscoring their potential to reliably enhance the performance of a wide range of MR models. Our results demonstrate that the inclusion of MedChat and HyperMed can reliably improve the performance of most MR models.

4) RQ4: Analysis of Historical Visits: In MR tasks, effectively utilizing patients' historical visit records is pivotal for simulating the diagnostic process of physicians and provides essential support for decision-making. We further investigate the impact of the length of historical visit records used by the framework (k) on performance. This analysis not only seeks in enhancing the predictive accuracy but also helps in finding an optimal balance between leveraging comprehensive historical data and maintaining computational efficiency. As shown in Figure 6, our analysis within the MIMIC-III dataset reveals that the performance of MEDICS improved as k increased from 1 to 3, but slightly declined when k extends to 4 and 5. In the MIMIC-IV dataset, MEDICS demonstrates optimal performance at k was 4. The observed decline in performance with longer historical spans can be attributed to the inclusion of potentially outdated or less relevant data, which may compromise the accuracy of the predictions. To mitigate these

issues, we recommend selecting a moderate length of historical records. This approach helps balance the benefits of accessing comprehensive historical data with the need to maintain model efficiency and data relevance.

5) Case study: Table V presents a case study analysis. The “Correct” column lists the number of medications correctly identified using the target electronic health records (EHR) data and our recommendations. The “Missed” column indicates the medications in the target EHR data that our recommendations failed to identify, while the “Unseen” column includes medications recommended by the system MEDICS that were absent in the target EHR data.

For this study, we selected a patient from the test set with multiple visits to assess the accuracy of medication combination predictions across different models. The results indicate that, during the patient’s first visit, all models correctly predicted the same number of medications; however, MEDICS and SafeDrug identified the fewest medications not listed in the EHR data. In the second visit, MEDICS surpassed other models by correctly predicting a greater number of medications and reducing the occurrence of unseen medications. This illustrates the effectiveness of our framework in leveraging historical patient data and closely aligning with physician prescribing patterns to enhance medication prediction accuracy.

V. DISCUSSION

A. Clinical Relevance

Medication recommendation is essential for clinical decision-making, particularly when treating patients with multifaceted conditions and medication histories. The MEDICS framework enhances this task by integrating multi-source domain knowledge and modeling both semantic and structural information, thus aligning more closely with physicians’ cognitive workflows. This leads to improved personalization of treatment plans, reduced medication errors, and optimized health outcomes.

In clinical practice, where patient records span multiple visits and involve complex temporal patterns, MEDICS emulates the reasoning process of clinicians by incorporating longitudinal data and knowledge-grounded inferences. Furthermore, the use of foundation model components, such as MedChat, offers the flexibility to support diverse clinical documentation styles and coding systems, increasing the framework’s potential adaptability to different healthcare environments.

While the model demonstrates strong empirical performance, real-world deployment would also require attention to explainability, clinician trust, and infrastructure compatibility. These considerations, though beyond the scope of this study, are crucial for translating algorithmic improvements into clinical impact.

B. Innovation

The core innovation of MEDICS lies in its dual-perspective integration of semantic and structural domain knowledge for medication recommendation—a contrast to traditional methods that primarily rely on flat tabular data. By leveraging multi-source descriptions and hierarchical classifications from

external medical ontologies, MEDICS uncovers multi-modal and multi-view representations that better reflect the clinical context.

Specifically, MedChat redefines the MR task as a temporal conversational problem, aligning more naturally with the physician’s diagnostic process. HyperMed complements this by capturing high-order correlations among clinical events via sub-hypergraph modeling. Together, these modules not only improve performance but also simulate reasoning patterns that are more interpretable and aligned with clinical logic.

Another key advantage is modularity: both MedChat and HyperMed are designed to be plug-in components that can augment a variety of existing MR models. Our compatibility analysis confirms that these modules consistently enhance multiple baseline methods, demonstrating MEDICS’ value not only as a standalone framework but also as a transferable enhancement toolkit for broader clinical prediction scenarios.

C. Limitations and Future Work

While MEDICS demonstrates promising performance across multiple clinical prediction tasks, several areas offer opportunities for further improvement and extension.

1) Cross-system and Clinical Language Adaptability: MEDICS was evaluated on MIMIC-III and MIMIC-IV, both derived from U.S.-based intensive care units. Differences in healthcare systems—including drug availability, coding practices, and disease prevalence—may affect its generalizability to other regions. Moreover, the semantic encoder within HyperMed currently leverages PubMedBERT, which is well aligned with formal biomedical knowledge sources but may not optimally capture the linguistic nuances of clinical narratives. Future work should explore external validation on diverse datasets and evaluate alternative encoders, such as ClinicalBERT [15] or UMLSBERT [29], particularly for tasks involving unstructured clinical text.

2) Enhancing Clinical Relevance and Interpretability: The current evaluation pipeline primarily relies on retrospective data and quantitative metrics such as Jaccard score, PRAUC, and DDI rate. No clinician-in-the-loop assessment has yet been performed, which limits insight into the clinical plausibility of model recommendations. Additionally, although the framework integrates semantic and structural signals via LLMs and hypergraphs, the interpretability of predictions remains limited from a clinician’s perspective. Future research should incorporate expert validation through prospective studies or structured scoring protocols and explore explainability techniques such as attention visualization, rule extraction from the hypergraph, or LLM-generated rationales to facilitate real-world adoption.

3) Scalability and Prompt Optimization: MEDICS integrates large-scale LLM inference (MedChat), hypergraph learning (HyperMed), and collaborative training, resulting in non-trivial resource demands. Although LoRA was applied for efficient fine-tuning, deployment in low-resource settings may require further optimization via model distillation or modular pruning. Moreover, the summarization prompt in the KnowRetr module was manually crafted through empirical tuning, lacking formal ablation or automated refinement. Future work should explore data-driven prompt engineering methods such as prompt

tuning or instruction learning to improve consistency and reproducibility.

4) Broadening the Scope of Drug Interaction Modeling:

Currently, the framework focuses primarily on minimizing antagonistic drug-drug interactions. Synergistic or neutral combinations are not explicitly modeled, nor are pharmacodynamic or pharmacokinetic mechanisms incorporated. Integrating pharmacological knowledge graphs and modeling a broader spectrum of drug interactions may enhance the clinical relevance and safety of medication recommendations.

VI. CONCLUSION

In this work, we introduced MEDICS, a novel MR framework that harnesses both structural and semantic insights from EHR data, enriched with multi-source domain knowledge. We designed three plugin modules tailored for clinical prediction tasks: *KnowRetr* enables multi-source knowledge retrieval and summarization for effective information utilization. *MedChat* utilizes LLMs for accurate clinical predictions. *HyperMed* integrates structural and semantic node embeddings through sub-hypergraph encoding to capture complex interrelationships. Furthermore, a co-training strategy is employed to enhance the model's robustness and generalization ability. Experiments confirm the accuracy of the MEDICS and show that our plugin modules can be orthogonally applied to and improve other existing MR models, indicating broad applicability. The distinct strength and versatility of MEDICS stem from its capability to integrate continuously updated domain knowledge and adapt various pre-trained models for specific tasks.

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